Lecture Scribe Probability

L4_A.pdf
PDF

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material.

CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel

Purpose: Exam-oriented lecture scribe faithful to Lecture 4 slides.

This scribe follows the lecture structure exactly and develops each concept step-by-step so that a student can reconstruct the reasoning used in class during revision. Only material appearing in the lecture slides is used.

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1. Introduction to Probability Theory

1.1 Experiment, Outcome, Event, and Sample Space

Probability theory begins with formalizing random procedures.

Definition: Experiment (E)
An **experiment** is a procedure that produces a result.

Example from lecture:

- Tossing a coin five times.

Each execution produces a sequence of heads (H) and tails (T).

Definition: Outcome (ω)
An **outcome** is one possible result of an experiment.

Example:
If five coins are tossed, one possible outcome is:

$$\omega = HHTHT$$

Each full sequence corresponds to one atomic result.

Definition: Event
An **event** is a set of outcomes.

Example:
Event C = outcomes containing an even number of heads.

Thus:

- An event is not a single outcome,
- It is a collection of outcomes satisfying a condition.

Definition: Sample Space (S)
The **sample space** is the set of all possible distinct outcomes.

Properties:

- Outcomes are **mutually exclusive** (cannot occur simultaneously).
- Outcomes are **collectively exhaustive** (all possibilities included).

Example:
For one coin toss:

$$S = \{H, T\}$$

Sample spaces may be:

- Discrete,
- Countably infinite,
- Continuous.

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2. Axioms of Probability

Probability assigns numerical likelihoods to events.

Axiom 1 — Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

For sample space S,

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For many mutually exclusive events,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

These axioms are assumed true and require no proof.

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3. Corollaries and Propositions

From axioms we derive useful rules.

Proposition 1 — Complement Rule

$$Pr(A^c) = 1 - Pr(A)$$

Reasoning:

Total probability is 1, and event plus its complement covers all outcomes.

Proposition 2 — Monotonicity

If $A \subset B$,

$$Pr(A) \leq Pr(B)$$

Reason:

Event B contains all outcomes of A and possibly more.

Proposition 3 — Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Intersection must be subtracted once to avoid double counting.

Proposition 4 — General Inclusion–Exclusion

Probability of union of many events alternates sums and intersections.

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4. Assigning Probabilities

4.1 Classical Approach

When outcomes are equally likely,

$$Pr(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

Examples:

- Probability of head in fair coin = 1/2.
- Probability of even number in die roll = 3/6.

4.2 Relative Frequency Approach

Probability estimated by repeated trials:

$$Pr(A) \approx \frac{\text{number of occurrences of A}}{\text{number of trials}}$$

Observation:

- As number of trials increases, ratio stabilizes.
- Drawback: infinite repetition is impossible.

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5. Joint Probability

Motivation

Events may occur together and are not always mutually exclusive.

Thus, we study:

$$Pr(A \cap B)$$

or

$$Pr(A, B)$$

for simultaneous occurrence.

Calculation Methods

1. Express events in atomic outcomes.
2. Find common outcomes.
3. Sum their probabilities.

Or estimate via simultaneous frequency.

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6. Example 1 — Card Deck

Given:

- 52 cards.
- 13 per suit.

Events:

- A: red card.

- B: number card (Ace counted).
- C: heart card.

Individual probabilities

Red cards = 26.

$$Pr(A) = \frac{26}{52} = \frac{1}{2}$$

Number cards = 10 per suit.

$$Pr(B) = \frac{40}{52}$$

Hearts = 13.

$$Pr(C) = \frac{13}{52}$$

Joint probabilities

Red & number:

20 cards.

$$Pr(A, B) = \frac{20}{52}$$

Red & heart:

All hearts are red.

$$Pr(A, C) = \frac{13}{52}$$

Number & heart:

10 cards.

$$Pr(B, C) = \frac{10}{52}$$

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7. Example 2 — Costume Party

Tops:

- 3 T-shirts

- 1 cape

Total = 4

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants
- 4 boxers

Total = 6

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent.

Joint probability:

$$Pr = (1/4) \times (4/6) = 1/6$$

Correct option: C.

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8. Conditional Probability

Motivation

Occurrence of one event may affect another.

Definition

Probability of A given B occurred:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

$$Pr(A, B) = Pr(A|B)Pr(B) = Pr(B|A)Pr(A)$$

Chain Rule

For multiple events:

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

Generalizes for many events.

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9. Example 3 — Cards Without Replacement

First card: spade.

Remaining cards:

- Total = 51
- Spades = 12

Thus,

$$Pr(B|A) = \frac{12}{51}$$

Second draw probability changes due to removal.

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10. Example 4 — Poker Flush

A flush = all cards same suit.

Flush in Spades

Probabilities sequentially:

$$\frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} \times \frac{10}{49} \times \frac{9}{48}$$

Each step reduces cards and suit count.

Any Suit Flush

Four suits, mutually exclusive:

$$P(\text{Any Flush}) = 4 \times P(\text{Spade Flush})$$

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11. Example 5 — Missing Key

Given:

- Left pocket probability = 0.4
- Right pocket probability = 0.4
- Jacket total = 0.8

Search left pocket and key not found.

We compute:

$$P(R|L^c) = \frac{P(R)}{1 - P(L)}$$

Substituting:

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

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12. Key Exam Revision Points

Students should be able to:

✓ Define experiment, outcome, event, sample space.
✓ State probability axioms.
✓ Apply complement and union rules.
✓ Compute classical probabilities.
✓ Compute joint probabilities.
✓ Use conditional probability formula.
✓ Apply product rule and chain rule.
✓ Solve without-replacement problems.
✓ Compute poker flush probabilities.

If you'd like, next we can prepare a **one-page exam cheat sheet**, or **practice problems likely for exams**.

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CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-oriented lecture scribe prepared strictly from the provided lecture slides, structured for revision and faithful to lecture scope and sequencing.

This scribe reconstructs the logical development of concepts exactly as presented, so that students can follow definitions, derivations, and examples step-by-step during exam preparation. Only material present in the lecture slides is included.

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1. Foundations of Probability

Probability theory begins by formalizing experiments and outcomes.

1.1 Experiment

An **experiment** is any procedure that produces a result.

Example from lecture: tossing a coin five times. Each repetition of the procedure yields one result.

1.2 Outcome

An **outcome** is one possible result of an experiment.

For five coin tosses, an example outcome is:

$$\omega = HHTHT$$

Each sequence represents a single elementary result.

1.3 Event

An **event** is a collection (set) of outcomes satisfying some condition.

Example: event consisting of outcomes having an even number of heads.

Thus:

- An outcome is one result,
- An event is a set of such results.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Key properties:

- Outcomes are **mutually exclusive** (only one occurs per trial).
- Outcomes are **collectively exhaustive** (no possibility omitted).

Example for one coin toss:

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

---

2. Axioms of Probability

Probability values obey fundamental rules.

Axiom 1 – Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 – Total Probability

The probability of the sample space is:

$$Pr(S) = 1$$

Axiom 3 – Additivity

For mutually exclusive events,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

for mutually exclusive events.

These axioms are accepted as the foundation of probability theory.

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3. Derived Results (Corollaries & Propositions)

Using axioms, useful results are obtained.

3.1 Complement Rule

The complement A^c contains outcomes where A does not occur.

Since total probability equals 1:

$$Pr(A) + Pr(A^c) = 1$$

Therefore,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity

If event A is contained in event B,

$$A \subset B$$

then

$$Pr(A) \leq Pr(B)$$

because all outcomes in A also appear in B.

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection is subtracted once to avoid double counting.

3.4 Inclusion–Exclusion Extension

For multiple events, alternating sums and intersections appear.

---

4. Methods of Assigning Probability

4.1 Classical Method

When all outcomes are equally likely:

$$Pr(A) = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

Examples:

- Head in fair coin = 1/2.
- Even number in die roll = 3/6.

4.2 Relative Frequency Method

Probability approximated via repeated trials:

$$Pr(A) \approx \frac{\text{occurrences of A}}{\text{number of trials}}$$

As trials increase, values stabilize, though infinite repetition is impractical.

---

5. Joint Probability

Events need not be mutually exclusive. Sometimes events occur together.

Joint probability:

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B)$$

represents simultaneous occurrence.

Computation:

1. Express events using outcomes,
2. Identify common outcomes,
3. Sum their probabilities.

---

6. Example – Card Deck Probabilities

Given:

- 52-card deck,
- 13 cards per suit.

Define:

- A: card is red,
- B: card is a number card (Ace counted),
- C: card is a heart.

Individual probabilities

Red cards = 26:

$$Pr(A) = 26/52 = 1/2$$

Number cards = 10 per suit:

$$Pr(B) = 40/52$$

Hearts = 13:

$$Pr(C) = 13/52$$

Joint probabilities

Red and number cards:
20 cards → $20/52$

Red and hearts:
All hearts are red → $13/52$

Number cards and hearts:
10 cards → $10/52$.

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7. Example – Costume Party Selection

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent, so

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct option: C.

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8. Conditional Probability

When occurrence of one event influences another, conditional probability is used.

Definition:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition:

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and symmetrically,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events:

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

Generalizes to more events.

---

9. Example – Drawing Cards Without Replacement

First card is a spade.

Remaining cards:

- Total cards = 51,
- Spades remaining = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes because cards are not replaced.

---

10. Example – Poker Flush Probability

A flush requires all cards from one suit.

Flush in spades

Sequential selection:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Each draw reduces both total cards and suit count.

Any suit flush

Since four suits exist and events are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush})$$

---

11. Example – Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After not finding key in left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substituting:

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

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12. Exam Preparation Checklist

Students should be able to:

- Define experiment, outcome, event, sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

This completes the lecture scribe faithful to Lecture 4 content and structure.

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CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides, preserving structure, sequencing, and reasoning as presented in class.

This document follows the lecture content step by step. Definitions, logical reasoning, derivations, and example solutions are written explicitly so students can reproduce the reasoning process during exam revision. Only material contained in the provided lecture slides is used.

---

1. Foundations of Probability

Probability theory begins by formally describing experiments and their possible outcomes.

1.1 Experiment

An **experiment** is defined as a procedure that produces a result.

Lecture example: tossing a coin five times. Each performance of the procedure yields one result sequence.

1.2 Outcome

An **outcome** is one possible result of the experiment.

For five coin tosses, a possible outcome is:

$$\omega = HHTHT$$

Each full sequence corresponds to one atomic result.

1.3 Event

An **event** is a set of outcomes satisfying some condition.

Example from lecture: the event consisting of outcomes containing an even number of heads.

Hence:

- Outcome → single result.
- Event → set of outcomes meeting a condition.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Key properties:

- Outcomes are **mutually exclusive** (only one occurs).
- Outcomes are **collectively exhaustive** (all possibilities included).

Example:
For one coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

2. Axioms of Probability

Probability values obey three axioms.

Axiom 1 — Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

The probability of the entire sample space is:

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For many mutually exclusive events,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

These axioms serve as the foundation of probability theory.

---

3. Derived Results (Corollaries and Propositions)

From axioms, useful rules follow.

3.1 Complement Rule

Event complement A^c contains outcomes where A does not occur.

Since the total probability equals 1,

$$Pr(A) + Pr(A^c) = 1$$

Thus,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity Property

If

$$A \subset B$$

then event B contains all outcomes of A and possibly more, giving

$$Pr(A) \leq Pr(B)$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection is subtracted to correct double counting.

3.4 Inclusion–Exclusion Extension

For multiple events, alternating additions and subtractions of intersections are used.

---

4. Assigning Probabilities

4.1 Classical Method

When all outcomes are equally likely,

$$Pr(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

Lecture examples:

- Probability of head = 1/2.
- Probability of even number on a die = 3/6.

4.2 Relative Frequency Method

Probability estimated through repeated trials:

$$Pr(A) \approx \frac{\text{number of occurrences of A}}{\text{number of trials}}$$

The value stabilizes as trials increase, though infinite repetition is impossible.

---

5. Joint Probability

Events are not always mutually exclusive; they may occur together.

Joint probability is:

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B)$$

representing simultaneous occurrence.

Calculation steps:

1. Express events via outcomes.
2. Identify outcomes common to both events.
3. Sum their probabilities.

---

6. Example — Card Deck Probabilities

Given:

- Standard 52-card deck,
- 13 cards per suit.

Define events:

- A: card is red,
- B: card is a number card (Ace counted),
- C: card is a heart.

Individual probabilities

Red cards:

$$Pr(A) = 26/52 = 1/2$$

Number cards:

$$Pr(B) = 40/52$$

Hearts:

$$Pr(C) = 13/52$$

Joint probabilities

Red and number cards:
20 cards,

$$Pr(A, B) = 20/52$$

Red and hearts:
all hearts are red,

$$Pr(A, C) = 13/52$$

Number cards and hearts:
10 cards,

$$Pr(B, C) = 10/52$$

---

7. Example — Costume Party Selection

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct answer: Option C.

---

8. Conditional Probability

When one event influences another, conditional probability is used.

Definition:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition,

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and symmetrically,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events:

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

Extends similarly to additional events.

---

9. Example — Cards Drawn Without Replacement

First card drawn is a spade.

Remaining cards:

- Total cards = 51,
- Remaining spades = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes because the first card is not replaced.

---

10. Example — Poker Flush Probability

A flush requires all cards to belong to one suit.

Flush in spades

Sequential probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Counts decrease after each draw.

Any Suit Flush

Since suits are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush})$$

---

11. Example — Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After failing to find the key in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substitution gives:

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

---

12. Exam Revision Checklist

Students must be able to:

- Define experiment, outcome, event, and sample space.
- State and apply probability axioms.
- Use complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

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CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and accompanying textbook excerpts. The structure, sequencing, derivations, and example reasoning follow the lecture material faithfully, enabling step-by-step revision for examinations.

This scribe presents each concept with explicit reasoning exactly in the logical order developed in the lecture so that students can reconstruct arguments and calculations during exam preparation. No external material or additional assumptions are introduced.

---

1. Foundations of Probability

Probability theory begins by formally describing experiments and their possible outcomes.

1.1 Experiment

An **experiment** is a procedure that produces a result when performed.

Lecture example: tossing a coin five times. Each performance yields one complete sequence of results.

1.2 Outcome

An **outcome** is one possible result of the experiment.

For five coin tosses, a sample outcome is:

$$\omega = HHTHT$$

Each full sequence corresponds to one atomic result.

1.3 Event

An **event** is a set of outcomes satisfying some condition.

Example: event consisting of outcomes containing an even number of heads.

Thus,

- Outcome → single result.

- Event → set of outcomes satisfying a property.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Properties:

- Outcomes are **mutually exclusive**, meaning only one outcome occurs per experiment.
- Outcomes are **collectively exhaustive**, meaning all possibilities are included.

Example:
For one coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

---

2. Axioms of Probability

Probability assignments obey three axioms.

Axiom 1 — Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

The probability of the entire sample space equals one:

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

for mutually exclusive events.

These axioms are taken as foundational rules.

---

3. Results Derived from the Axioms

Using the axioms, important consequences follow.

3.1 Complement Rule

The complement A^c contains outcomes where A does not occur.

Since probabilities of an event and its complement together cover the sample space:

$$Pr(A) + Pr(A^c) = 1$$

Thus,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity

If

$$A \subset B$$

then all outcomes of A are also in B, implying

$$Pr(A) \leq Pr(B)$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection is subtracted because it is counted twice in the sum.

3.4 Inclusion–Exclusion Extension

For several events, alternating addition and subtraction of intersections yields union probabilities.

---

4. Methods for Assigning Probabilities

4.1 Classical Method

If all outcomes are equally likely,

$$Pr(A) = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

Examples:

- Probability of head = 1/2.
- Probability of even number on a die = 3/6.

4.2 Relative Frequency Method

Probability is approximated by repeated experiments:

$$Pr(A) \approx \frac{\text{number of times A occurs}}{\text{number of trials}}$$

As trials increase, results stabilize, although infinite repetition is impossible.

---

5. Joint Probability

Events need not be mutually exclusive; they may occur simultaneously.

Joint probability:

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B)$$

represents simultaneous occurrence.

Computation procedure:

1. Express events in terms of outcomes.
2. Identify common outcomes.

3. Sum probabilities of common outcomes.

---

6. Example — Card Deck Probabilities

Given:

- Standard deck: 52 cards.
- 13 cards per suit.

Define:

- A: card is red,
- B: card is a number card (Ace included),
- C: card is a heart.

Individual probabilities

Red cards = 26:

$$Pr(A) = 26/52 = 1/2$$

Number cards = 40:

$$Pr(B) = 40/52$$

Hearts = 13:

$$Pr(C) = 13/52$$

Joint probabilities

Red and number cards:
20 cards,

$$Pr(A, B) = 20/52$$

Red and hearts:
all hearts are red,

$$Pr(A, C) = 13/52$$

Number cards and hearts:
10 cards,

$$Pr(B, C) = 10/52$$

---

7. Example — Costume Party Selection

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct answer: Option C.

---

8. Conditional Probability

When occurrence of one event affects another, conditional probability applies.

Definition:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition,

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and also,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

Extends similarly for more events.

---

9. Example — Cards Without Replacement

First card drawn is a spade.

Remaining cards:

- Total cards = 51,
- Spades remaining = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes since the first card is not replaced.

---

10. Example — Poker Flush Probability

A flush requires all cards from one suit.

Flush in spades

Sequential probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Counts reduce after each draw.

Any Suit Flush

Since suits are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush})$$

---

11. Example — Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After not finding the key in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substitution:

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

---

12. Exam Revision Checklist

Students must be able to:

- Define experiment, outcome, event, and sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

This scribe reproduces Lecture 4 content faithfully and provides a structured exam-reference document strictly based on the provided lecture material.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material.

CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and relevant textbook excerpts. The structure, definitions, derivations, and examples follow the lecture exactly so that students can reproduce reasoning during revision.

This scribe presents each concept in the logical order developed in the lecture. Definitions, assumptions, and example solutions are explained step by step exactly as presented, without introducing external material.

---

1. Foundations of Probability

Probability theory begins with formalizing experiments and possible outcomes.

1.1 Experiment

An **experiment** is a procedure that produces a result.

Lecture example: tossing a coin five times. Each execution of the procedure generates one result sequence.

1.2 Outcome

An **outcome** is one possible result of the experiment.

For five coin tosses, an outcome may be:

$$\omega = HHTHT$$

Each complete sequence corresponds to a single elementary result.

1.3 Event

An **event** is a set of outcomes satisfying a certain condition.

Lecture example: outcomes containing an even number of heads.

Thus:

- Outcome → single result.
- Event → collection of outcomes meeting a condition.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Properties:

- Outcomes are **mutually exclusive** (only one occurs).
- Outcomes are **collectively exhaustive** (all possibilities included).

Example:
For one coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

---

2. Axioms of Probability

Probability assignments obey three axioms.

Axiom 1 – Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 – Total Probability

The probability of the entire sample space equals:

$$Pr(S) = 1$$

Axiom 3 – Additivity

For mutually exclusive events,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

when events are mutually exclusive.

These axioms form the basis of probability theory.

---

3. Results Derived from Axioms

Using axioms, important properties follow.

3.1 Complement Rule

The complement A^c contains outcomes where A does not occur.

Since event and complement together form the sample space,

$$Pr(A) + Pr(A^c) = 1$$

Therefore,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity Property

If

$$A \subset B$$

then all outcomes of A lie within B, so

$$Pr(A) \leq Pr(B)$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection is subtracted to avoid double counting.

3.4 Inclusion–Exclusion Principle

For multiple events, alternating sums and intersections are used to compute union probabilities.

---

4. Assigning Probabilities

4.1 Classical Method

When outcomes are equally likely,

$$Pr(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

Lecture examples:

- Probability of head = 1/2.
- Probability of even number on a die = 3/6.

4.2 Relative Frequency Method

Probability can be approximated through repeated experiments:

$$Pr(A) \approx \frac{\text{number of times A occurs}}{\text{number of trials}}$$

The value stabilizes as trials increase, although infinite repetition is impractical.

---

5. Joint Probability

Events may occur simultaneously and are not always mutually exclusive.

Joint probability is written as:

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B)$$

representing both events occurring together.

Computation steps:

1. Represent events using outcomes.
2. Identify outcomes common to both events.
3. Sum their probabilities.

---

6. Example – Card Deck Probabilities

Given:

- Standard deck: 52 cards.
- 13 cards per suit.

Define events:

- A: card is red,
- B: card is a number card (Ace included),
- C: card is a heart.

Individual probabilities

Red cards:

$$Pr(A) = 26/52 = 1/2$$

Number cards:

$$Pr(B) = 40/52$$

Hearts:

$$Pr(C) = 13/52$$

Joint probabilities

Red and number cards:
20 cards,

$$Pr(A, B) = 20/52$$

Red and hearts:
all hearts are red,

$$Pr(A, C) = 13/52$$

Number cards and hearts:
10 cards,

$$Pr(B, C) = 10/52$$

---

7. Example – Costume Party Selection

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct answer: Option C.

8. Conditional Probability

When occurrence of one event affects another, conditional probability is defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition,

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and similarly,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

This extends similarly for more events.

---

9. Example – Cards Drawn Without Replacement

If the first card drawn is a spade:

Remaining cards:

- Total cards = 51,
- Remaining spades = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes since the card is not replaced.

---

10. Example – Poker Flush Probability

A flush requires all cards from the same suit.

Flush in spades

Sequential probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Counts decrease after each draw.

Any Suit Flush

Since suits are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush})$$

---

11. Example – Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After not finding the key in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substituting values,

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

---

12. Exam Revision Checklist

Students must be able to:

- Define experiment, outcome, event, and sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

This completes the lecture scribe faithfully reproducing Lecture 4 content and reasoning, serving as an exam-ready reference strictly based on the provided lecture material.

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CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel

Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and relevant textbook excerpts. The presentation follows the lecture structure and

reasoning exactly so that students can reproduce arguments and calculations during revision.

This scribe develops each concept step by step, maintaining the logical flow used in the lecture. Only material contained in the provided context is included; no external assumptions or examples are introduced.

---

1. Foundations of Probability

Probability theory begins with formal definitions of experiments and their outcomes.

1.1 Experiment

An **experiment** is a procedure that produces a result.

Lecture example: tossing a coin five times. Each execution of the procedure generates one sequence of results.

1.2 Outcome

An **outcome** is one possible result of the experiment.

For five coin tosses, a sample outcome is:

$$\omega = HHTHT$$

Each sequence represents one elementary result.

1.3 Event

An **event** is a set of outcomes satisfying a specified condition.

Example: outcomes containing an even number of heads.

Thus,

- Outcome → one result.
- Event → a collection of outcomes satisfying a rule.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Properties:

- Outcomes are **mutually exclusive** (only one occurs in an experiment).
- Outcomes are **collectively exhaustive** (all possibilities included).

Example:
For one coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

---

2. Axioms of Probability

Probability assignments obey three axioms.

Axiom 1 — Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

The probability of the entire sample space equals one:

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

when events are mutually exclusive.

These axioms form the foundation of probability theory.

---

3. Results Derived from the Axioms

From the axioms, several useful properties follow.

3.1 Complement Rule

The complement A^c contains outcomes where A does not occur.

Since event and complement together equal the sample space,

$$Pr(A) + Pr(A^c) = 1$$

Thus,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity Property

If

$$A \subset B,$$

then event B contains all outcomes of A, implying

$$Pr(A) \leq Pr(B).$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B).$$

The intersection is subtracted to avoid double counting.

3.4 Inclusion–Exclusion Principle

For multiple events, union probabilities are computed by alternating addition and subtraction of intersections.

---

4. Assigning Probabilities

4.1 Classical Method

If all outcomes are equally likely,

$$Pr(A) = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}.$$

Lecture examples:

- Probability of head in a fair coin = $1/2$.
- Probability of even number in a die roll = $3/6$.

4.2 Relative Frequency Method

Probability can be approximated through repeated experiments:

$$Pr(A) \approx \frac{\text{number of occurrences of A}}{\text{number of trials}}.$$

The value stabilizes as trials increase, although infinite repetition is impossible.

---

5. Joint Probability

Events are not always mutually exclusive; they may occur together.

Joint probability is written as

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B),$$

representing simultaneous occurrence.

Computation steps:

1. Express events in terms of outcomes.
2. Identify outcomes common to both events.
3. Sum probabilities of common outcomes.

---

6. Example — Card Deck Probabilities

Given:

- Standard deck: 52 cards.
- 13 cards per suit.

Events:

- A: card is red,
- B: card is a number card (Ace included),
- C: card is a heart.

Individual probabilities

Red cards:

$$Pr(A) = 26/52 = 1/2.$$

Number cards:

$$Pr(B) = 40/52.$$

Hearts:

$$Pr(C) = 13/52.$$

Joint probabilities

Red and number cards:

$$Pr(A, B) = 20/52.$$

Red and hearts:

$$Pr(A, C) = 13/52.$$

Number cards and hearts:

$$Pr(B, C) = 10/52.$$

---

7. Example — Costume Party Selection

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4.$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6.$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6.$$

Correct answer: Option C.

---

8. Conditional Probability

When occurrence of one event influences another, conditional probability is defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0.$$

Product Rule

From the definition,

$$Pr(A, B) = Pr(A|B)Pr(B),$$

and also,

$$Pr(A, B) = Pr(B|A)Pr(A).$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A).$$

This extends similarly to additional events.

---

9. Example — Cards Drawn Without Replacement

If the first card drawn is a spade:

Remaining cards:

- Total cards = 51,
- Remaining spades = 12.

Thus,

$$Pr(B|A) = 12/51.$$

Probability changes because the first card is not replaced.

---

10. Example — Poker Flush Probability

A flush requires all cards from the same suit.

Flush in spades

Sequential probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}.$$

Counts decrease after each draw.

Any suit flush

Since suits are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush}).$$

---

11. Example — Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After not finding the key in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}.$$

Substituting,

$$= \frac{0.4}{0.6} = \frac{2}{3}.$$

Correct answer: Option C.

---

12. Exam Revision Checklist

Students must be able to:

- Define experiment, outcome, event, and sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

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CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel

Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and relevant textbook excerpts. The structure, reasoning, derivations, and examples follow the lecture exactly, enabling students to reproduce the logical development during revision.

This scribe presents concepts in the same logical order as the lecture. Definitions, assumptions, derivations, and example solutions are written step by step so that the reasoning process used in class is preserved. No additional material or assumptions beyond the provided context are introduced.

---

1. Foundations of Probability

Probability theory begins by formally defining experiments and their outcomes.

1.1 Experiment

An **experiment** is a procedure that produces a result when performed.

Lecture example: tossing a coin five times. Each execution produces one sequence of outcomes.

1.2 Outcome

An **outcome** is one possible result of an experiment.

For five coin tosses, one possible outcome is:

$$\omega = HHTHT$$

Each complete sequence represents one elementary result.

1.3 Event

An **event** is a set of outcomes satisfying a given condition.

Example used in lecture: outcomes containing an even number of heads.

Thus:

- An outcome is a single result,
- An event is a collection of outcomes satisfying a condition.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Properties:

- Outcomes are **mutually exclusive**, meaning only one outcome occurs per experiment.
- Outcomes are **collectively exhaustive**, meaning all possible outcomes are included.

Example:
For one coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

---

2. Axioms of Probability

Probability assignments follow three axioms.

Axiom 1 — Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

The probability of the sample space equals:

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

when events are mutually exclusive.

These axioms form the foundation of probability theory.

---

3. Results Derived from the Axioms

From the axioms, important consequences follow.

3.1 Complement Rule

The complement A^c contains outcomes where A does not occur.

Since event and complement together form the sample space,

$$Pr(A) + Pr(A^c) = 1$$

Hence,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity Property

If

$$A \subset B,$$

then all outcomes of A are contained in B, implying

$$Pr(A) \leq Pr(B)$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection is subtracted to avoid double counting.

3.4 Inclusion–Exclusion Principle

For multiple events, union probabilities are computed using alternating additions and subtractions of intersections.

---

4. Assigning Probabilities

4.1 Classical Method

If all outcomes are equally likely,

$$Pr(A) = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

Lecture examples:

- Probability of head in a fair coin = $1/2$.
- Probability of even number in a die roll = $3/6$.

4.2 Relative Frequency Method

Probability may be approximated through repeated experiments:

$$Pr(A) \approx \frac{\text{occurrences of A}}{\text{number of trials}}$$

As trials increase, the ratio stabilizes, although infinite repetition is impossible.

---

5. Joint Probability

Events are not always mutually exclusive and may occur together.

Joint probability is written as

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B)$$

representing simultaneous occurrence.

Computation steps:

1. Express events using outcomes.
2. Identify outcomes common to both events.
3. Sum probabilities of these outcomes.

---

6. Example — Card Deck Probabilities

Given:

- Standard deck of 52 cards,
- 13 cards per suit.

Define:

- A: card is red,
- B: card is a number card (Ace counted),
- C: card is a heart.

Individual probabilities

Red cards:

$$Pr(A) = 26/52 = 1/2$$

Number cards:

$$Pr(B) = 40/52$$

Hearts:

$$Pr(C) = 13/52$$

Joint probabilities

Red and number cards:
20 cards,

$$Pr(A, B) = 20/52$$

Red and hearts:
all hearts are red,

$$Pr(A, C) = 13/52$$

Number cards and hearts:

10 cards,

$$Pr(B, C) = 10/52$$

---

7. Example — Costume Party Selection

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct answer: Option C.

---

8. Conditional Probability

When occurrence of one event affects another, conditional probability is defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition,

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and also,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

This extends similarly for additional events.

---

9. Example — Cards Drawn Without Replacement

If the first card drawn is a spade:

Remaining cards:

- Total cards = 51,
- Remaining spades = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes since the first card is not replaced.

---

10. Example — Poker Flush Probability

A flush requires all cards from one suit.

Flush in spades

Sequential probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Counts decrease after each draw.

Any suit flush

Since suits are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush})$$

---

11. Example — Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After not finding the key in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substituting,

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

---

12. Exam Revision Checklist

Students must be able to:

- Define experiment, outcome, event, and sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

This completes the lecture scribe, faithfully reproducing Lecture 4 content and reasoning to serve as exam reference material based strictly on the provided lecture context.

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CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and relevant textbook excerpts. The presentation preserves the lecture's structure, sequencing, and reasoning so students can follow and reproduce each step during examination revision.

This scribe develops definitions, assumptions, derivations, and example solutions exactly in the logical order presented in the lecture. Explicit reasoning steps are retained so that the progression from definitions to results is clear. No additional assumptions, explanations, or examples beyond the provided context are introduced.

---

1. Foundations of Probability

Probability theory begins with formal descriptions of experiments and their possible outcomes.

1.1 Experiment

An **experiment** is a procedure that produces a result when performed.

Lecture illustration: tossing a coin five times. Each performance produces one sequence of outcomes.

1.2 Outcome

An **outcome** is one possible result of an experiment.

For five coin tosses, an outcome may be:

$$\omega = HHTHT$$

Each sequence corresponds to one elementary result.

1.3 Event

An **event** is a set of outcomes satisfying a specified condition.

Lecture example: the event consisting of outcomes having an even number of heads.

Thus,

- Outcome refers to one result,
- Event refers to a collection of outcomes satisfying a condition.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes of an experiment.

Properties emphasized in lecture:

- Outcomes are **mutually exclusive**, meaning only one occurs per experiment.
- Outcomes are **collectively exhaustive**, meaning all possibilities are included.

Example:
For a single coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

---

2. Axioms of Probability

Probability values satisfy three axioms.

Axiom 1 — Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

The probability of the sample space equals one:

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

for mutually exclusive events.

These axioms form the foundation from which further results follow.

---

3. Results Derived from the Axioms

From the axioms, useful properties are derived.

3.1 Complement Rule

The complement A^c contains outcomes where A does not occur.

Since event and complement together form the entire sample space,

$$Pr(A) + Pr(A^c) = 1$$

Hence,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity Property

If

$$A \subset B,$$

then event B contains all outcomes of A, so

$$Pr(A) \leq Pr(B)$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection is subtracted because it is counted twice in the sum.

3.4 Inclusion–Exclusion Principle

For multiple events, union probabilities are obtained using alternating additions and subtractions of event intersections.

---

4. Assigning Probabilities

4.1 Classical Method

When outcomes are equally likely,

$$Pr(A) = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

Lecture examples:

- Probability of head in a fair coin is $1/2$.
- Probability of even number in a die roll is $3/6$.

4.2 Relative Frequency Method

Probability may be estimated using repeated trials:

$$Pr(A) \approx \frac{\text{number of occurrences of A}}{\text{number of trials}}$$

The ratio stabilizes as trials increase, though infinite repetition is impossible.

---

5. Joint Probability

Events need not be mutually exclusive and may occur simultaneously.

Joint probability is expressed as:

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B)$$

representing simultaneous occurrence.

Procedure used in lecture:

1. Express events in terms of outcomes.
2. Identify outcomes common to both events.
3. Sum probabilities of these outcomes.

---

6. Example — Card Deck Probabilities

Given:

- Standard deck with 52 cards,
- 13 cards per suit.

Events:

- A: card is red,
- B: card is a number card (Ace included),
- C: card is a heart.

Individual probabilities

Red cards:

$$Pr(A) = 26/52 = 1/2$$

Number cards:

$$Pr(B) = 40/52$$

Hearts:

$$Pr(C) = 13/52$$

Joint probabilities

Red and number cards:

$$Pr(A, B) = 20/52$$

Red and hearts:

$$Pr(A, C) = 13/52$$

Number cards and hearts:

$$Pr(B, C) = 10/52$$

---

7. Example — Costume Party Selection

Tops available:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms available:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent, so

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct answer: Option C.

---

8. Conditional Probability

When one event influences another, conditional probability is defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition,

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and equivalently,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

This extends similarly for additional events.

---

9. Example — Cards Drawn Without Replacement

If the first card drawn is a spade:

Remaining cards:

- Total cards = 51,
- Remaining spades = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes because the first card is not replaced.

---

10. Example — Poker Flush Probability

A flush requires all cards from one suit.

Flush in spades

Sequential probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Counts reduce after each draw.

Any suit flush

Since suits are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush})$$

---

11. Example — Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After not finding the key in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substituting values,

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

---

12. Exam Revision Checklist

Students must be able to:

- Define experiment, outcome, event, and sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

This completes the lecture scribe faithfully reproducing Lecture 4 material as an exam reference document, strictly following the provided lecture content.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and relevant textbook excerpts. The structure, definitions, derivations, and worked examples follow the lecture exactly so that students can reproduce reasoning during examination revision.

This scribe preserves the lecture's logical development. Each concept is presented in sequence, with explicit reasoning steps used in class. Only material contained in the provided lecture context is included; no additional assumptions or examples are introduced.

---

1. Foundations of Probability

Probability theory begins by formally defining experiments and their possible outcomes.

1.1 Experiment

An **experiment** is a procedure that produces a result when performed.

Lecture example: tossing a coin five times. Each execution produces one sequence of outcomes.

1.2 Outcome

An **outcome** is one possible result of an experiment.

For five coin tosses, one possible outcome is:

$$\omega = HHTHT$$

Each complete sequence corresponds to one elementary result.

1.3 Event

An **event** is a set of outcomes satisfying a specified condition.

Lecture example: outcomes containing an even number of heads.

Thus,

- Outcome refers to one result,
- Event refers to a collection of outcomes satisfying a rule.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Key properties:

- Outcomes are **mutually exclusive**, meaning only one occurs per experiment.
- Outcomes are **collectively exhaustive**, meaning all possible outcomes are included.

Example:
For one coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete, countably infinite, or continuous.

---

2. Axioms of Probability

Probability assignments satisfy three axioms.

Axiom 1 — Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

The probability of the sample space equals:

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

for mutually exclusive events.

These axioms serve as the foundation of probability theory.

---

3. Results Derived from the Axioms

Using the axioms, further properties are derived.

3.1 Complement Rule

The complement A^c contains outcomes where A does not occur.

Since event and complement together form the sample space,

$$Pr(A) + Pr(A^c) = 1$$

Therefore,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity Property

If

$$A \subset B,$$

then all outcomes of A lie within B, implying

$$Pr(A) \leq Pr(B)$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection is subtracted to avoid double counting.

3.4 Inclusion–Exclusion Principle

For several events, probabilities of unions are computed using alternating sums and intersections.

---

4. Assigning Probabilities

4.1 Classical Method

When outcomes are equally likely,

$$Pr(A) = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

Lecture examples:

- Probability of head in a fair coin = $1/2$.
- Probability of even number on a die = $3/6$.

4.2 Relative Frequency Method

Probability may be approximated using repeated trials:

$$Pr(A) \approx \frac{\text{number of occurrences of A}}{\text{number of trials}}$$

As trials increase, the ratio stabilizes, although infinite repetition is not possible.

---

5. Joint Probability

Events may occur simultaneously and are not always mutually exclusive.

Joint probability is written as:

$$Pr(A \cap B) \quad \text{or} \quad Pr(A, B)$$

representing simultaneous occurrence.

Lecture computation procedure:

1. Represent events via outcomes.
2. Identify outcomes common to both events.
3. Sum their probabilities.

---

6. Example — Card Deck Probabilities

Given:

- Standard deck with 52 cards,
- 13 cards per suit.

Define:

- A: card is red,
- B: card is a number card (Ace included),
- C: card is a heart.

Individual probabilities

Red cards:

$$Pr(A) = 26/52 = 1/2$$

Number cards:

$$Pr(B) = 40/52$$

Hearts:

$$Pr(C) = 13/52$$

Joint probabilities

Red and number cards:

$$Pr(A, B) = 20/52$$

Red and hearts:

$$Pr(A, C) = 13/52$$

Number cards and hearts:

$$Pr(B, C) = 10/52$$

---

7. Example — Costume Party Selection

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent, therefore

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct answer: Option C.

---

8. Conditional Probability

When occurrence of one event influences another, conditional probability is defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition,

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and equivalently,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

This extends similarly to additional events.

---

9. Example — Cards Drawn Without Replacement

If the first card drawn is a spade:

Remaining cards:

- Total cards = 51,
- Remaining spades = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes since the card is not replaced.

---

10. Example — Poker Flush Probability

A flush requires all cards from the same suit.

Flush in spades

Sequential probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Counts decrease after each draw.

Any suit flush

Since suits are mutually exclusive,

$$Pr(\text{flush}) = 4 \times Pr(\text{spade flush})$$

---

11. Example — Missing Key Problem

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After not finding the key in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substituting,

$$= \frac{0.4}{0.6} = \frac{2}{3}$$

Correct answer: Option C.

---

12. Exam Revision Checklist

Students must be able to:

- Define experiment, outcome, event, and sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.

- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

This completes the lecture scribe, faithfully reproducing Lecture 4 content and reasoning to serve as exam reference material strictly based on the provided lecture material.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and textbook excerpts. The structure, definitions, derivations, and worked examples follow the lecture sequence so students can reproduce results reliably during revision.

This document adheres strictly to the lecture scope and presents definitions, assumptions, derivations, and example solutions as given in the lecture material. No external examples or assumptions are introduced.

---

1. Basic Terminology of Probability

Experiment

An **experiment** is a procedure that produces an observable result.

Example used in lecture: tossing a coin five times.

Outcome

An **outcome** is one possible result of the experiment.

Example:

$$\omega = HHTHT$$

Each sequence is one outcome.

Event

An **event** is a collection of outcomes satisfying a condition.

Lecture example: outcomes containing an even number of heads.

Sample Space

The **sample space**, denoted S, is the set of all outcomes.

Properties emphasized:

- Outcomes are **mutually exclusive**.
- Outcomes are **collectively exhaustive**.

Example:

$$S = \{H, T\}$$

for one coin toss.

Sample spaces may be discrete or continuous.

---

2. Axioms of Probability

Probability assignments satisfy:

Axiom 1: Non-negativity

$$0 \leq Pr(A) \leq 1$$

Axiom 2: Total Probability

$$Pr(S) = 1$$

Axiom 3: Additivity

For mutually exclusive events,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For mutually exclusive collections,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

---

3. Properties Derived from Axioms

Complement Rule

Since event and complement form the sample space,

$$Pr(A^c) = 1 - Pr(A)$$

Monotonicity

If

$$A \subset B,$$

then

$$Pr(A) \leq Pr(B)$$

Union Formula

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Intersection is subtracted to avoid double counting.

Inclusion–Exclusion Principle

Union probability for several events is computed using alternating sums of intersections.

---

4. Assigning Probabilities

Classical Method

When outcomes are equally likely,

$$Pr(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

Examples:

- Fair coin: $Pr(H) = 1/2$
- Even number on die: $3/6$

Relative Frequency Method

Probability estimated as

$$Pr(A) \approx \frac{\text{occurrences of } A}{\text{number of trials}}$$

Ratio stabilizes as trials increase.

---

5. Joint Probability

Joint probability represents simultaneous occurrence:

$$Pr(A \cap B)$$

Computation steps used:

1. Express events via outcomes.
2. Identify common outcomes.
3. Sum probabilities of common outcomes.

---

6. Card Deck Example

Deck: 52 cards, 13 per suit.

Define:

- A: red card
- B: number card (Ace included)
- C: heart

Individual probabilities:

$$Pr(A) = 26/52$$

$$Pr(B) = 40/52$$

$$Pr(C) = 13/52$$

Joint probabilities:

$$Pr(A, B) = 20/52$$

$$Pr(A, C) = 13/52$$

$$Pr(B, C) = 10/52$$

---

7. Costume Party Example

Tops: 3 T-shirts + 1 cape → total 4

$$Pr(\text{Cape}) = 1/4$$

Bottoms: 2 pants + 4 boxers → total 6

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct option: C.

---

8. Conditional Probability

Defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

$$Pr(A, B) = Pr(A|B)Pr(B)$$

Also,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

---

9. Cards Without Replacement

If first card is spade:

Remaining:

- Total cards = 51
- Spades left = 12

Thus,

$$Pr(B|A) = 12/51$$

Probability changes due to removal.

---

10. Poker Flush Probability

Flush requires all cards of one suit.

Spade flush probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Any suit flush:

$$4 \times \text{spade flush probability}$$

---

11. Missing Key Example

Given:

- Left pocket: 0.4
- Right pocket: 0.4
- Jacket: 0.8

After failing in left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

So,

$$= \frac{0.4}{0.6} = 2/3$$

Correct option: C.

---

12. Exam Preparation Checklist

Students should be able to:

- Define experiment, outcome, event, sample space.
- State and apply probability axioms.
- Use complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

End of Lecture 4 Scribe — faithful to lecture material and structured for exam revision.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel
Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and textbook excerpts. The document preserves the lecture's structure, reasoning sequence, and worked examples so students can follow each logical step during revision and reproduce results in examinations.

All definitions, assumptions, derivations, and example solutions are presented in the order and scope of the lecture material. No external explanations or additional assumptions are introduced.

---

1. Basic Terminology of Probability

1.1 Experiment

An **experiment** is a procedure that produces a result when performed.

Lecture illustration: tossing a coin five times, where each trial produces one sequence of outcomes.

1.2 Outcome

An **outcome** is a single possible result of the experiment.

Example:

$$\omega = HHTHT$$

Each sequence of five tosses is one outcome.

1.3 Event

An **event** is a set of outcomes satisfying a specified condition.

Lecture example: outcomes containing an even number of heads.

Thus:

- Outcome → one result.
- Event → collection of outcomes satisfying a rule.

1.4 Sample Space

The **sample space**, denoted S, is the set of all possible outcomes.

Lecture notes emphasize:

- Outcomes are **mutually exclusive**, meaning only one occurs per experiment.
- Outcomes are **collectively exhaustive**, meaning all possible results are included.

Example:
For a single coin toss,

$$S = \{H, T\}$$

Sample spaces may be discrete or continuous.

---

2. Axioms of Probability

Probability values satisfy three axioms.

Axiom 1: Non-negativity

For any event A,

$$0 \leq Pr(A) \leq 1$$

Axiom 2: Total Probability

Probability of the sample space equals one:

$$Pr(S) = 1$$

Axiom 3: Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally, for mutually exclusive events,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

These axioms form the basis for all subsequent results.

---

3. Properties Derived from Axioms

3.1 Complement Rule

Event and complement together form the sample space:

$$Pr(A) + Pr(A^c) = 1$$

Thus,

$$Pr(A^c) = 1 - Pr(A)$$

3.2 Monotonicity Property

If

$$A \subset B,$$

then event B contains all outcomes of A, giving

$$Pr(A) \leq Pr(B)$$

3.3 Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection term is subtracted because it is counted twice.

3.4 Inclusion–Exclusion Principle

For multiple events, union probabilities are computed using alternating sums of intersections, following lecture derivation.

---

4. Assigning Probabilities

4.1 Classical Method

If outcomes are equally likely,

$$Pr(A) = \frac{\text{number of favorable outcomes}}{\text{total outcomes}}$$

Lecture examples:

- Fair coin: $Pr(H) = 1/2$
- Even number on die: $3/6$

4.2 Relative Frequency Method

Probability may be estimated through repeated trials:

$$Pr(A) \approx \frac{\text{occurrences of } A}{\text{total trials}}$$

The ratio stabilizes as trials increase.

---

5. Joint Probability

Joint probability represents simultaneous occurrence:

$$Pr(A \cap B) \text{ or } Pr(A, B)$$

Lecture computation steps:

1. Express events in terms of outcomes.
2. Identify outcomes common to both events.
3. Sum probabilities of those outcomes.

---

6. Card Deck Example

Given:

- 52 cards,
- 13 cards per suit.

Define events:

- A: card is red,
- B: card is a number card (Ace included),
- C: card is a heart.

Individual probabilities:

$$Pr(A) = 26/52$$

$$Pr(B) = 40/52$$

$$Pr(C) = 13/52$$

Joint probabilities:

$$Pr(A, B) = 20/52$$

$$Pr(A, C) = 13/52$$

$$Pr(B, C) = 10/52$$

---

7. Costume Party Example

Tops available:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms available:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct option: C.

---

8. Conditional Probability

Defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

From the definition:

$$Pr(A, B) = Pr(A|B)Pr(B)$$

and also,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events:

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

---

9. Cards Drawn Without Replacement

If first card is a spade:

Remaining cards:

- Total cards = 51,
- Remaining spades = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes because cards are not replaced.

---

10. Poker Flush Probability

Flush requires all cards from one suit.

Probability for spade flush:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Probability for any suit:

$$4 \times \text{spade flush probability}$$

---

11. Missing Key Example

Given probabilities:

- Left pocket = 0.4,
- Right pocket = 0.4,
- Jacket = 0.8.

After not finding key in left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Substituting:

$$= \frac{0.4}{0.6} = 2/3$$

Correct answer: Option C.

---

12. Revision Checklist

Students should be able to:

- Define experiment, outcome, event, and sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

End of Lecture 4 Scribe — structured precisely for exam revision based solely on the provided lecture material.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material.

CSE400 – Fundamentals of Probability in Computing

Lecture 4 Scribe: Joint Probability and Conditional Probability

Instructor: Dr. Dhaval Patel

Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides and textbook excerpts. The structure, definitions, derivations, and worked examples follow

the lecture sequence so students can reproduce results reliably during examination revision.

All content below adheres strictly to the lecture scope. Definitions, assumptions, derivations, and example solutions are presented in the order given in the lecture material. No additional material or assumptions are introduced.

---

1. Basic Terminology of Probability

Experiment

An **experiment** is a procedure that produces a result when performed.

Lecture example: tossing a coin five times, where each performance produces one sequence of outcomes.

Outcome

An **outcome** is a single possible result of the experiment.

Example outcome:

$$\omega = HHTHT$$

Each sequence corresponds to one outcome.

Event

An **event** is a set of outcomes satisfying a specified condition.

Lecture example: outcomes containing an even number of heads.

Hence:

- Outcome = one result,
- Event = collection of outcomes satisfying a rule.

Sample Space

The **sample space** S is the set of all possible outcomes.

Properties emphasized:

- Outcomes are mutually exclusive.
- Outcomes are collectively exhaustive.

Example:

$$S = \{H, T\}$$

for a single coin toss.

Sample spaces may be discrete or continuous.

---

2. Axioms of Probability

Probability assignments satisfy:

Axiom 1 — Non-negativity

$$0 \leq Pr(A) \leq 1$$

Axiom 2 — Total Probability

$$Pr(S) = 1$$

Axiom 3 — Additivity

If events are mutually exclusive,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

More generally,

$$Pr\left(\bigcup_i A_i\right) = \sum_i Pr(A_i)$$

for mutually exclusive events.

---

3. Properties Derived from the Axioms

Complement Rule

Since event and complement form the sample space,

$$Pr(A) + Pr(A^c) = 1$$

Thus,

$$Pr(A^c) = 1 - Pr(A)$$

Monotonicity Property

If

$$A \subset B,$$

then

$$Pr(A) \leq Pr(B)$$

Union Rule

For any events,

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

The intersection term avoids double counting.

Inclusion–Exclusion Principle

Union probabilities for multiple events are computed using alternating sums of intersections, as shown in lecture.

---

4. Assigning Probabilities

Classical Method

When outcomes are equally likely,

$$Pr(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

Lecture examples:

- Fair coin: $Pr(H) = 1/2$
- Even number on die: $3/6$

Relative Frequency Method

Probability estimated through repeated trials:

$$Pr(A) \approx \frac{\text{occurrences of } A}{\text{number of trials}}$$

The ratio stabilizes as trials increase.

---

5. Joint Probability

Joint probability represents simultaneous occurrence:

$$Pr(A \cap B)$$

Lecture computation steps:

1. Express events as sets of outcomes.
2. Identify outcomes common to both events.
3. Sum probabilities of common outcomes.

---

6. Card Deck Example

Deck properties:

- Total cards = 52,
- Each suit = 13 cards.

Events defined:

- A: card is red,
- B: card is a number card (Ace included),
- C: card is a heart.

Individual probabilities:

$$Pr(A) = 26/52$$

$$Pr(B) = 40/52$$

$$Pr(C) = 13/52$$

Joint probabilities:

$$Pr(A, B) = 20/52$$

$$Pr(A, C) = 13/52$$

$$Pr(B, C) = 10/52$$

---

7. Costume Party Example

Tops:

- 3 T-shirts,
- 1 cape → total 4.

$$Pr(\text{Cape}) = 1/4$$

Bottoms:

- 2 pants,
- 4 boxers → total 6.

$$Pr(\text{Boxers}) = 4/6$$

Selections are independent:

$$Pr(\text{Cape and Boxers}) = (1/4)(4/6) = 1/6$$

Correct answer: Option C.

---

8. Conditional Probability

Defined as:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}, \quad Pr(B) > 0$$

Product Rule

$$Pr(A, B) = Pr(A|B)Pr(B)$$

Also,

$$Pr(A, B) = Pr(B|A)Pr(A)$$

Chain Rule

For three events,

$$Pr(A, B, C) = Pr(C|A, B)Pr(B|A)Pr(A)$$

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9. Cards Drawn Without Replacement

If the first card is a spade:

Remaining:

- Total cards = 51,
- Spades left = 12.

Thus,

$$Pr(B|A) = 12/51$$

Probability changes because cards are not replaced.

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10. Poker Flush Probability

Flush requires all cards of one suit.

Spade flush probability:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Any suit flush:

$$4 \times \text{spade flush probability}$$

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11. Missing Key Example

Given:

- Left pocket probability = 0.4,
- Right pocket probability = 0.4,
- Jacket probability = 0.8.

After failing in the left pocket:

$$Pr(R|L^c) = \frac{Pr(R)}{1 - Pr(L)}$$

Thus,

$$= \frac{0.4}{0.6} = 2/3$$

Correct answer: Option C.

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12. Exam Preparation Checklist

Students should be able to:

- Define experiment, outcome, event, sample space.
- State probability axioms.
- Apply complement and union rules.
- Compute classical probabilities.
- Compute joint probabilities.
- Apply conditional probability.
- Use product and chain rules.
- Solve without-replacement problems.
- Compute flush probabilities.

End of Lecture 4 Scribe — prepared strictly according to the provided lecture material for examination revision.